Focused Vortex Core Efficiency at a Predicted Geometric Ring Crossing: A Bounded Scalar Angular-Spectrum Study

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Abstract

We examine a model-authored question for focused $p = 0$ vortices passed through an ideal thin lens of focal length $f = 0.06$ m: does a fixed on-axis core of radius $r_c = 5 \times 10^{-5}$ m show a step-like loss of core efficiency when the focused vortex ring crosses the core edge, and do width scaling, anisotropic stretch, and exact algebraic nulls distinguish this geometric account from a broadened-diffraction account? The zero-parameter ring-crossing prediction is $l^* = r_c\, 2\pi w/(\lambda f)$, about 2.48 for $w = 3.0\times10^{-4}$ m and 1.49 for the intended $w = 1.8\times10^{-4}$ m. In the executed scalar angular-spectrum runs, the measured core efficiency falls from 0.33710 at charge $l = 2$ to 0.15325 at $l = 3$ for $w = 3.0 \times 10^{-4}$ m, a factor 2.20, close to the predicted adjacent-charge crossing. However, the intended width-shift arm was not executed at the smaller width: the recorded executed waist for that arm is still 3.0×10^{-4} m, and its metrics are identical to the $l = 2$ reference. A single-charge anisotropic stretch gives 0.15263, a ratio of 0.996 to the isotropic $l = 3$ value, and cannot locate a step. The 180° rotation null passes exactly in the supplied arms, while the charge-sign reversal null is not present. Optimized classical controls reach core efficiencies 0.8771, 0.8761, and 0.8825, comparable to the 0.8706 Gaussian anchor, so the fixed-core observable alone is not specific to a vortex ring crossing. The $l = 2$ to $l = 3$ drop is a reproducible numerical fact in this run and is consistent with the geometric ring-crossing estimate, but the available discriminators do not establish the mechanism.

1 Introduction

Focused structured light provides a way to convert a known input phase profile into a controlled focal distribution. For scalar paraxial fields, a thin lens maps an input vortex of charge l into an annular focal structure whose radius scales with l, wavelength, and focal length, and inversely with the input beam size. When the focal distribution is measured through a fixed circular aperture, the measured fraction can depend on whether the annular ring lies inside or outside that aperture. This paper asks whether such a fixed-core measurement shows a sharp, parameter-free step at the charge where the focused ring crosses the core radius, and whether controlled changes in width and symmetry separate this geometric explanation from a generic charge-dependent diffraction or broadening explanation.

The question addressed here is model-authored and was registered with a fixed observable, core efficiency, before the numerical runs. It is not an axial-floor experiment, and the inputs are Gaussian or $p = 0$ vortex fields; no Airy, Pearcey, or Butterfly mask is used in this article. The central prediction is geometric. A focused $p = 0$ vortex of charge l is expected to form a ring of radius

$$\rho_l = \frac{l\lambda f}{2\pi w}, \tag{1}$$

where w is the input Gaussian envelope radius and f is the focal length. A fixed on-axis core of radius r_c intersects the ring while $\rho_l < r_c$, and misses it when $\rho_l > r_c$. The crossing charge is therefore

$$l^* = \frac{r_c\, 2\pi w}{\lambda f}. \tag{2}$$

For $r_c = 5\times10^{-5}$ m, $\lambda = 6.33\times10^{-7}$ m, $f = 0.06$ m, and $w = 3.0\times10^{-4}$ m, Eq. (2) gives $l^* = 2.48$. Charge $l = 2$ has $\rho_2 = 4.03 \times 10^{-5}$ m, inside the core, while $l = 3$ has $\rho_3 = 6.04 \times 10^{-5}$ m, outside the core. If the mechanism is radial geometry, the fixed-core efficiency should drop between these adjacent charges. A width reduction to $w = 1.8 \times 10^{-4}$ m would move l^* to 1.49, so the same $l = 2$ input should fall on the low side of the step. An anisotropic stretch should break the circular ring symmetry and either blur or shift the step. Exact algebraic nulls, namely charge-sign reversal and a 180° rotation, provide run-validity checks but are not by themselves mechanism evidence.

The executed experiments provide one clear numerical step and several informative negative or invalid checks. At $w = 3.0 \times 10^{-4}$ m, core efficiency falls from 0.33710 at $l = 2$ to 0.15325 at $l = 3$, a factor of 2.20. The intended width-shift arm, however, was recorded with the same executed waist 3.0×10^{-4} m as the reference and reproduces its metrics exactly; it is therefore not a valid width test. The stretch arm at a single charge gives a ratio of 0.996, and the charge-sign null is absent. The present article reports these outcomes as bounded numerical findings and does not upgrade the adjacent-charge drop into a confirmed geometric mechanism.

2 Related Work

Structured scalar beams with engineered phase and amplitude are routinely propagated through free space and focused by lenses. Laguerre–Gaussian and vortex fields carry orbital angular momentum and have a long history in scalar beam theory [2, 1]. More general phase-structured families, including catastrophe beams, have been studied for autofocusing, self-healing, and controlled caustic formation [5, 11, 9, 8, 10]. Those works motivate the use of phase-structured inputs and focal-plane diagnostics, but they do not supply the present numerical evidence or the specific fixed-core crossing test.

Numerical propagation of such fields commonly uses angular-spectrum or Fresnel methods. Band-limited angular-spectrum propagation is standard for free-space scalar fields [4]. Here the propagator is a scalar angular-spectrum method with a conservative radial bandwidth cutoff. This is sufficient for the registered comparison, but it is not the complete Matsushima construction. Structured-light diagnostics also include mode sorting and phase-only reshaping [3, 7]; the present observable is narrower, namely the fraction of power falling inside a fixed circular core at each propagation plane, and the interpretation is limited accordingly.

3 Theoretical model

The scalar input on the numerical grid is normalized to power P,

$$U_0(x_j, y_k) = \sqrt{\frac{P}{\Delta x^2 \sum_{a,b} |V_{ab}|^2}}\, V(x_j, y_k), \tag{3}$$

where V is the sampled input profile and Δx is the grid spacing. For the Gaussian anchor, the executed family is `gaussian` with scale w. For the vortex arms, the executed family is `vortex` with charge l and scale w. The supplied records identify the input by these fields and by the normalized profile in Eq. (3). No Airy, Pearcey, or Butterfly input is used, and the unused Butterfly controls are zero. The recorded apodization value 0.15 applies only to Airy inputs in this system, so it does not modify the Gaussian or vortex profiles used here. The active inputs are therefore set by w and l alone.

For a thin lens of focal length f, the lens phase $\exp[-i\pi(x^2 + y^2)/(\lambda f)]$ is applied to the input before propagation. The propagated field at distance z is

$$U_z = \mathcal{F}^{-1}\Big[\mathcal{F}(U_0)\, B_z \exp\Big(i2\pi z\sqrt{\lambda^{-2} - f_x^2 - f_y^2}\Big)\Big], \tag{4}$$

where f_x, f_y are spatial frequencies, $\mathcal{F}$ is the two-dimensional Fourier transform, and B_z is a conservative radial cutoff: $B_z = 1$ for $\sqrt{f_x^2 + f_y^2} \le f_{\max}(z)$ and $B_z = 0$ otherwise. In the lens runs, U_0 in Eq. (4) denotes the lens-modulated input. The cutoff suppresses evanescent and poorly sampled high-frequency components; it is not a full band-limited angular-spectrum construction.

Composition of two coherent arms uses an explicit rotation θ, relative phase δ, and amplitude weight a,

$$U_c = \sqrt{\frac{P}{\int |U + ae^{i\delta}\mathcal{R}_\theta U|^2 \, dA}} \, (U + ae^{i\delta}\mathcal{R}_\theta U), \tag{5}$$

where $\mathcal{R}_\theta$ rotates the field by θ. An exactly zero coherent field cannot be normalized; in that case other arms remain measurable and the composition arm is left undefined. In the present runs the executed composition is $a = 1.0$, $\delta = 0$, and $\theta = 0$ for every case, so the composition arm coincides with the base arm rather than forming a distinct superposition.

The primary observable is the fixed-core efficiency. At each plane,

$$C(z) = P^{-1} \int_{x^2+y^2 \le r_c^2} |U_z|^2 \, dA, \qquad \overline{C} = \mathrm{median}_{z \in \mathcal{Z}} \, C(z), \tag{6}$$

with $r_c = 5 \times 10^{-5}$ m and $\mathcal{Z}$ the set of propagation planes. The quantity $\overline{C}$ is the registered core efficiency. It is a fraction of the propagated power inside the fixed core, not an absolute concentration and not a negated optimizer score.

4 Numerical methods

All reported runs use a scalar angular-spectrum propagator with a thin lens applied. The executed optical setup is $n = 192$, grid extent 2.4×10^{-3} m, wavelength 6.33×10^{-7} m, input waist 3.0×10^{-4} m for the reference cases, focal length 0.06 m, and power 1.0 W. The propagation window is $z = 0.049$ to 0.071 m in 21 planes. The recorded environment is `thin_lens` with phase noise rms 0.02, and the recorded composition is angle 0, phase 0, weight 1.0. The lens flag is true for every arm. The ridge diagnostic uses a marginal half width of 6×10^{-4} m and prominence threshold 0.25.

Five principal arms were executed. H001 is the $l = 0$ Gaussian anchor at $w = 3.0 \times 10^{-4}$ m. H002 is a vortex with $l = 2$ at $w = 3.0 \times 10^{-4}$ m. H003 is a vortex with $l = 3$ at $w = 3.0 \times 10^{-4}$ m. H004 was intended as a vortex with $l = 2$ at $w = 1.8 \times 10^{-4}$ m. H005 is a vortex with $l = 3$, $w = 3.0 \times 10^{-4}$ m, and an anisotropic stretch of 1.3. The executed setup recorded for H004, however, still lists `waist_m`=3.0×10^{-4} m. Its core efficiency, axial floor, ridge separation, and sidelobe fraction are numerically identical to H002. The intended width change did not execute, so H004 is treated as an implementation or registry defect, not as a width-shift test.

Each arm also records a phase-noise perturbed arm with rms 0.02 and a seed in the range 101–105. The core efficiencies in Table 1 are the base/single-arm values. The exact 180° rotation null is represented by the rotated arm. The coherent and incoherent arms are recorded separately; with $a = 1$, $\delta = 0$, and $\theta = 0$, the coherent arm reduces to the same field as the base arm. An exactly zero coherent field did not occur in these runs.

Confirmation records were supplied for H001 and H002 under mesh, window, propagator, quantization, wavelength, and axial-sampling variations. These records show what actually ran. A diagnostic that fails a sampling or edge check is not treated as positive evidence. Table 2 lists the core efficiencies from those confirmation runs. The H001 and H002 mesh256 and mesh384 values differ by about 0.5% and 2.8%, respectively. At $n = 384$, the ASM and Fresnel core efficiencies agree to about 0.03% for both H001 and H002. Changing the window to $n = 576$ and increasing the axial sampling to 41 planes leave the core efficiencies essentially unchanged. The H002 quantization record gives 0.32321 and flags `sampling_retention=false`; its edge fraction

Table 1: Executed principal arms and the registered fixed-core efficiency $\overline{C}$. H004 was requested at $w = 1.8 \times 10^{-4}$ m but the executed setup reports $w = 3.0 \times 10^{-4}$ m and reproduces H002. Values are from the supplied evidence packet.

ID	input	w (m)	l	transformation	$\overline{C}$	sidelobe
H001	Gaussian	3.0×10^{-4}	0	none	0.87064	0.12936
H002	vortex	3.0×10^{-4}	2	none	0.33710	0.66290
H003	vortex	3.0×10^{-4}	3	none	0.15325	0.84675
H004	vortex	1.8×10^{-4} requested executed 3.0×10^{-4}	2	none	0.33710	0.66290
H005	vortex	3.0×10^{-4}	3	stretch 1.3	0.15263	0.84737

Table 2: Confirmation records for the H001 Gaussian anchor and H002 $l = 2$ vortex. The H002 quantization record fails the sampling-retention check and is listed only to preserve what ran.

condition	H001 $\overline{C}$	H002 $\overline{C}$	note
mesh256	0.87716	0.34976	ASM, $n = 256$
mesh384	0.87261	0.34036	ASM, $n = 384$
fresnel384	0.87259	0.34025	Fresnel, $n = 384$
window576	0.87261	0.34036	ASM, extent 3.6×10^{-3} m
quantization	0.87261	0.32321	H002 sampling check false
wavelength640	0.86945	0.33378	$\lambda = 6.4 \times 10^{-7}$ m
axial41	0.87261	0.34036	41 z-planes

is 1.24×10^{-3}, above the 10^{-3} gate, so it is not used as evidence. No corresponding mesh, Fresnel, or window confirmation records are present for H003, H004, or H005 in the supplied packet.

5 Results and physical interpretation

5.1 Calibration anchor and the adjacent-charge step

The $l = 0$ Gaussian anchor H001 gives $\overline{C} = 0.87064$ with sidelobe fraction 0.12936 and no ridge separation. The packet does not include the pre-registered diffraction-limited band, so this value is used as a numerical reference rather than scored as a passed calibration gate. It confirms that the fixed-core integral returns a high fraction for a compact focal spot under the same lens, power, aperture, and grid.

At $w = 3.0 \times 10^{-4}$ m, the vortex arms give a clear adjacent-charge drop. H002 ($l = 2$) has $\overline{C} = 0.33710$, sidelobe fraction 0.66290, ridge separation 7.5×10^{-5} m, and peak intensity 0.27080. H003 ($l = 3$) has $\overline{C} = 0.15325$, sidelobe fraction 0.84675, ridge separation 1.0×10^{-4} m, and peak intensity 0.22402. The ratio is

$$\frac{\overline{C}_{l=2}}{\overline{C}_{l=3}} = \frac{0.33710}{0.15325} = 2.20.$$

This is the main positive numerical observation. It is close to the zero-parameter geometric expectation $l^* = 2.48$: $l = 2$ lies below the crossing and $l = 3$ lies above it. The corresponding ring radii from Eq. (1) are 4.03×10^{-5} m and 6.04×10^{-5} m, on either side of the 5×10^{-5} m core. The simultaneous increase in sidelobe fraction and ridge separation is consistent with the annular focal structure moving outward relative to the fixed core.

The interpretation must remain bounded. A fixed-core fraction can fall when power moves from the core into an annular ring, but it can also fall through charge-dependent diffraction, sampling, or apodization. The adjacent-charge drop alone does not separate these possibilities. The intended discriminators are therefore essential, and in the present packet they are either invalid or incomplete.

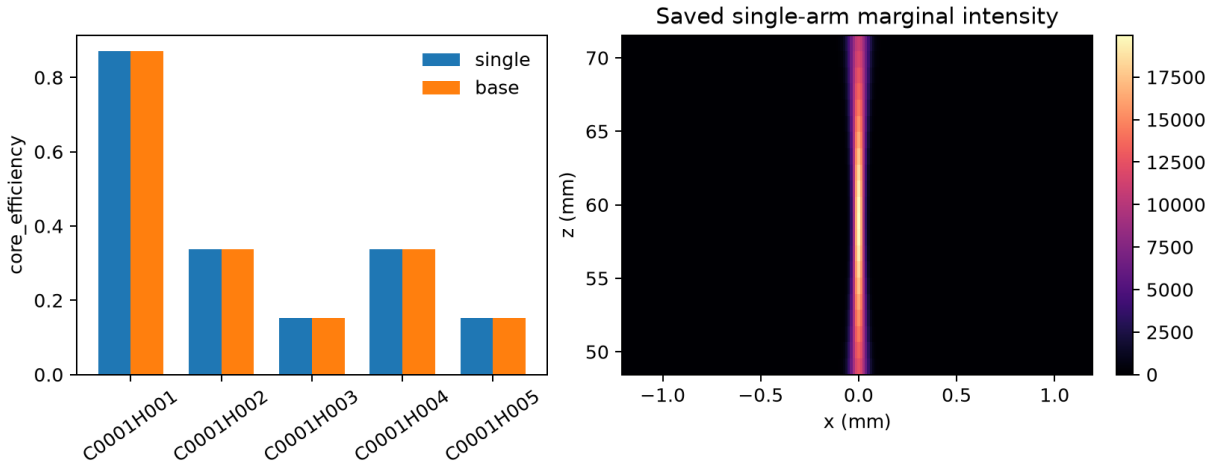


Figure 1: Overview of the executed scalar angular-spectrum results. The figure summarizes the registered core efficiencies and case labels; all quantitative statements in the text are based on the tabulated values in Tables 1 and 2. The figure is a summary, not an independent measurement.

5.2 Width-shift arm

The width-shift prediction is that reducing w from 3.0×10^{-4} m to 1.8×10^{-4} m moves l^* from 2.48 to 1.49, so the same $l = 2$ input should cross from the high side to the low side of the step. The supplied record does not execute that change. Although the case registry requests a scale of 1.8×10^{-4} m, the executed setup for H004 reports `waist_m`=3.0×10^{-4} m. Its metrics are identical to H002: $\overline{C} = 0.33710$, axial floor 3.65×10^{-36}, ridge separation 7.5×10^{-5} m, and sidelobe fraction 0.66290. The intended one-unit shift in l^* is therefore not observed, but the reason is that the input width was not changed in execution. This arm is not evidence for or against the geometric account. It identifies a registry or implementation defect that must be repaired before the width prediction can be tested.

5.3 Anisotropic stretch

The stretch arm H005 applies a factor 1.3 at $w = 3.0 \times 10^{-4}$ m and $l = 3$. The executed beam is visibly changed: the focal plane moves from 0.0589 m in H003 to 0.0556 m in H005, the peak intensity increases from 0.22402 to 0.29704, and the overlap with the unstretched base field is 0.8919. The registered core efficiency, however, is 0.15263, compared with 0.15325 for H003. The ratio is

$$\frac{\overline{C}_{\text{stretch}}}{\overline{C}_{\text{isotropic}}} = \frac{0.15263}{0.15325} = 0.996.$$

The ridge separation remains 7.5×10^{-5} m, while the isotropic $l = 3$ ridge separation is 1.0×10^{-4} m. A single charge point cannot determine a step location, so this arm cannot test the prediction that the step blurs or shifts by more than 0.5 charge units. What can be said is that the measured core efficiency at $l = 3$ is nearly unchanged by the stretch, and no step-location shift is measured. The stretch discriminator therefore does not support the geometric account as stated.

5.4 Nulls and classical controls

The 180° rotation null passes exactly in the supplied arms. For H001–H004, the rotated-arm core efficiency equals the base value to machine precision; for H005, the rotated value equals the transformed single-arm value 0.15263. This is a useful implementation check, but it is not a mechanism test. The charge-sign reversal null is not present in the packet. The sign-reversal comparison therefore remains untested, and no substitute conjugation argument is used here.

Optimized classical controls provide a further bound. With 32 attempts per family and the same power, aperture, lens, and grid, the best defocus control has $q_2 = -0.300$, $q_3 = 0$, $q_4 = 0$, ripple 0, and $\overline{C} = 0.87711$. The best quadratic/quartic control has $q_2 = -0.600$, $q_3 = 0$, $q_4 = 0$, ripple 0, and $\overline{C} = 0.87613$. The best four-parameter control has $q_2 = -0.629$, $q_3 = 0.00714$, $q_4 = -3.30 \times 10^{-4}$, ripple -2.83×10^{-3}, and $\overline{C} = 0.88251$. These values are comparable to or above the Gaussian anchor 0.87064. They show that a high fixed-core fraction is not by itself a specific signature of a vortex ring crossing. The controls do not explain the $l = 2$ to $l = 3$ drop, but they prevent reading the anchor or the controls as absolute concentration scores.

5.5 Numerical confirmation and scope

The confirmation runs in Table 2 support the numerical stability of the H001 and H002 values under mesh, window, propagator, and axial-sampling changes. The H001 anchor varies from 0.86945 to 0.87716 over the tested conditions. The H002 value varies from 0.34025 to 0.34976 over the passing conditions, with the quantization record excluded because its sampling-retention check fails. The ASM and Fresnel values at $n = 384$ agree closely, and the $n = 576$ window check is essentially identical to the $n = 384$ ASM result. These checks reduce the likelihood that the H002 value is a grid or window artifact. They do not cover H003, H004, or H005, so the $l = 3$ value and the stretch comparison have a narrower numerical basis than the $l = 2$ value.

The physical picture suggested by the data is therefore conditional. The focused vortex is expected to form a ring whose radius grows with charge. A fixed core of radius 5×10^{-5} m samples the central region; when the ring moves outward, the core fraction falls and the sidelobe fraction rises. The measured H002-to-H003 drop, the increase in sidelobe fraction from 0.663 to 0.847, and the ridge separation change from 7.5×10^{-5} m to 1.0×10^{-4} m are all consistent with this picture. However, the width-shift test did not change the width, the stretch test is a single charge point, and the sign-reversal null is absent. The classical controls show that the observable is not uniquely tied to a vortex ring. The data thus support a bounded statement: at $w = 3.0 \times 10^{-4}$ m, the fixed-core efficiency drops by a factor 2.20 between $l = 2$ and $l = 3$, near the zero-parameter crossing estimate $l^* = 2.48$; the mechanism remains unconfirmed.

6 Conclusions

This article reported a scalar angular-spectrum study of focused $p = 0$ vortices through an ideal $f = 0.06$ m thin lens with a fixed core radius $r_c = 5 \times 10^{-5}$ m. The zero-parameter geometric prediction places the ring-crossing charge at $l^* = 2.48$ for $w = 3.0 \times 10^{-4}$ m and $l^* = 1.49$ for the intended $w = 1.8 \times 10^{-4}$ m. The executed reference arms show a drop in core efficiency from 0.33710 at $l = 2$ to 0.15325 at $l = 3$, a factor of 2.20. The accompanying increase in sidelobe fraction from 0.66290 to 0.84675 and in ridge separation from 7.5×10^{-5} m to 1.0×10^{-4} m is consistent with the ring moving outward relative to the fixed core.

The discriminating tests do not confirm the geometric mechanism. The intended width-shift arm was executed with the same waist 3.0×10^{-4} m as the reference and reproduces its metrics exactly; it is a registry or implementation defect, not a width test. The anisotropic stretch at a single charge gives a core-efficiency ratio of 0.996 and no measured step shift. The 180° rotation null passes exactly, but the charge-sign reversal null is missing. Classical defocus, quadratic/quartic, and four-parameter controls reach 0.8771, 0.8761, and 0.8825, comparable to the 0.8706 Gaussian anchor, so the fixed-core observable is not specific to a vortex ring crossing. Confirmation records support the H001 and H002 values under mesh, window, Fresnel, and axial-sampling changes, but no equivalent confirmation records are present for H003–H005.

The reproducible numerical fact in this run is the $l = 2$ to $l = 3$ core-efficiency drop at $w = 3.0 \times 10^{-4}$ m and its proximity to the geometric crossing estimate. Establishing whether that drop is caused by the predicted radial ring crossing rather than charge-dependent diffraction

or sampling requires a validated width parameter, a charge scan at two widths with direct ring-radius measurement, a multi-charge stretch test, the missing sign-reversal null, and the full numerical preflights for every admitted vortex arm.

Disclosures

This manuscript was produced by Auto BeamScientist, which adapts Sakana AI Scientist-v2 [6]. The author and PDF author are Auto BeamScientist Project. Numerical values are taken from the executed scalar angular-spectrum runs summarized in the supplied evidence packet.

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